

MS2106E

USB Analog Video Capture IC

English reference translation of the Chinese MS2106E Datasheet, Version 1.0 (January 2021)

Translation note

This is a technical English rendering of the source datasheet. Signal names, pin numbers, electrical limits, interface names, units, and stated default behavior are retained. It is a design reference, not a manufacturer-issued English datasheet. Verify the final schematic against the original document and a sample device.

OpenXTCM5 relevance

For the planned rear-camera path, the key claims are: CVBS/S-Video input; USB 2.0 UVC 1.0 video output; maximum advertised video output of 720 x 480 at 25 Hz; optional MJPEG; and external EEPROM support for configuration. The published operating range is 0 to +70 C, so this component remains a prototype/validation choice rather than an automotive-qualified production selection.

1. Overview

The MS2106E is a video and audio capture IC. It integrates a USB 2.0 controller and data transceiver, video ADC, audio ADC, and audio/video processing. It transfers CVBS, S-Video, and audio signals over USB for preview or capture on a PC, smartphone, or tablet.

It supports YUV422 and MJPEG video output formats and is described as compatible with Windows, Android, and macOS.

2. Features

Video input

- CVBS and S-Video inputs
- NTSC/PAL compatible
- Automatic input-source detection

Audio input

- Integrated 24-bit audio ADC
- External I2S digital-audio input supported

Video output

- MJPEG and YUV422 output modes
- Maximum output: 720 x 480 at 25 Hz
- User-configurable output resolutions
- Brightness, contrast, saturation, and hue adjustments
- USB Video Class (UVC) 1.0 compliant

Audio output

- 48 kHz stereo USB audio
- USB Audio Class (UAC) 1.0 compliant

USB and system

- Integrated USB 2.0 device
- Custom descriptors supported
- Listed operating-system support: Windows XP/7/8/10, Android 5.0 or later, and macOS
- Requires an external 12 MHz passive crystal; internal PLL
- Integrated 3.3 V and 1.2 V LDOs; internal power-on reset
- 48-pin, 7 mm x 7 mm LQFP; lead-free and RoHS compliant

3. Applications

Audio/video capture cards; security monitoring; medical imaging; automotive rear-view video; and industrial cameras.

4. Functional Description

The functional block diagram shows a video ADC receiving CVBS or S-Video, an audio ADC receiving analog audio, an audio/video processing path, UVC 1.0 and UAC 1.0 encoding, a USB 2.0 device, EEPROM, MCU, programmable digital I/O, PMU, and PLL. The stated maximum video output is 720 x 480 and the stated USB audio output is 48 kHz stereo.

4.1 GPIO

Signal	Default function
GPIO0	Audio input source select. Grounded: select external I2S input. Floating: select the internal audio ADC.
GPIO2 (SCL)	Defaults to external EEPROM SCL.
GPIO3 (SDA)	Defaults to external EEPROM SDA.
GPIO4	Defaults to external EEPROM write-protect (WP).

4.2 Audio Input

The IC has an integrated 24-bit high-performance audio ADC and also accepts external I2S digital audio. The default source is the internal audio ADC. Ground GPIO0 to select external I2S input.

4.3 Analog Video Input

The IC has four video inputs and can support up to two inputs simultaneously. It supports CVBS and S-Video and automatically identifies the input source. If CVBS and S-Video are both connected, the default selection is the source connected first. If both are connected before power-up, the default is the source identified before the last power-down. The selection can also be customized through the external EEPROM.

4.4 USB Video Output

USB video conforms to UVC 1.0. YUV422 and MJPEG output are supported. The stated maximum output resolution is 720 x 480 at 25 Hz. Default output modes are listed below; external EEPROM may customize them.

USB 2.0 output format	Default resolution and rate
YUV422	480 x 320 at 25/20/10/5 Hz; 320 x 240 at 25/20/10/5 Hz
MJPEG	720 x 480; 640 x 480; 480 x 320; and 320 x 240, each at 25/20/10/5 Hz

4.5 USB Audio Output

USB audio conforms to UAC 1.0. The default output is 48 kHz stereo.

5. Pin Descriptions - I/O and Signal Pins

Pin name	#	Type	Description
GPIO0	35	Digital I/O	Audio-source select: ground for I2S input; floating for internal audio ADC.
GPIO2 / SCL	13	Digital I/O	General digital I/O; defaults to external EEPROM SCL.
GPIO3 / SDA	12	Digital I/O	General digital I/O; defaults to external EEPROM SDA.
GPIO4	19	Digital I/O	General digital I/O; defaults to external EEPROM WP.
XTALO	26	Analog output	12 MHz crystal output.
XTALI	27	Analog input	12 MHz crystal input.
REXT	31	Analog output	Connect a 4.7 kOhm resistor to ground.
USBDP	32	Digital I/O	USB differential positive signal.
USBDM	33	Digital I/O	USB differential negative signal.
SVCIN	3	Analog input	S-Video chroma input.
SVYIN	6	Analog input	S-Video luma input.
AVIN	8	Analog input	CVBS input.
G1IN	5	Analog input	Analog video signal input.
I2S_SCLK	36	Digital input	I2S serial clock input.
I2S_MCLK	37	Digital output	12.288 MHz clock output.
I2S_SD0	38	Digital input	I2S SD0 input.
I2S_LRCLK	39	Digital input	I2S left/right-channel select input.
VCOML	41	Analog output	Left-channel common-mode output.
ALIN	42	Analog input	Left-channel audio input.
ARIN	43	Analog input	Right-channel audio input.
VCOMR	44	Analog output	Right-channel common-mode output.

6. Pin Descriptions - Power and Ground

Pin name	#	Type	Description
VDD5V	16	Power	5 V supply input.
VLDO33	17	Power	3.3 V LDO output.
VLDO12	18	Power	1.2 V LDO output.
DVDD12	24	Power	Digital 1.2 V supply.
AVDD12	48	Power	Analog supply, video ADC 1.2 V input.
DVDD33	10, 14, 20, 34, 46	Power	Digital 3.3 V supply.
AVDD33	1, 4, 7	Power	Analog supply, video ADC 3.3 V input.
AVDD33_AU	40	Power	Analog supply, audio ADC 3.3 V input.
AVDD33_USB	30	Power	Analog supply, USB 3.3 V input.
AVSS	2, 9, 15, 28, 29, 45	Ground	Analog ground.
DVSS	11, 21, 47	Ground	Digital ground.
NC	22, 23, 25	-	No connection required.

Pinout orientation

The package is an LQFP-48 viewed from the top. The original datasheet's pin diagram assigns pins 1 through 12 down the left side, 13 through 24 along the bottom from left to right, 25 through 36 up the right side, and 37 through 48 along the top from right to left.

Important power-architecture note

The datasheet identifies VLDO33 and VLDO12 as LDO outputs while also identifying multiple 3.3 V and 1.2 V supply pins. It does not, in this document, give an implementation diagram that unambiguously defines every external connection. For OpenXTCM5, follow a proven MS2106E reference design and validate startup, current draw, and rail sequencing on the bench before connecting it to the CM5 USB host.

7. Electrical Characteristics

7.1 Absolute Maximum Ratings

Parameter	Symbol	Rating	Unit
Maximum operating voltage	VDD5	5.5	V
Maximum operating voltage (AVDD33 and DVDD33)	VDD33	3.6	V
Maximum operating voltage (AVDD12 and DVDD12)	VDD12	1.3	V
Ambient operating temperature	TA	0 to +70	C
Storage temperature	Tsto	-55 to +125	C

Do not exceed the absolute-maximum ratings. Operation outside the specified range can permanently damage the device. Functional operation is guaranteed only within the conditions specified by the datasheet; operation at absolute-maximum conditions can affect reliability.

7.2 ESD Absolute Maximum Ratings

Model	Rating
Human Body Model (HBM)	plus/minus 2000 V
Machine Model (MM)	plus/minus 200 V
Charged Device Model (CDM)	plus/minus 500 V

The source document recommends appropriate ESD precautions. Despite dedicated ESD-protection circuitry, sufficiently energetic events can permanently damage the device.

7.3 DC Electrical Characteristics

Parameter	Minimum	Typical	Maximum	Unit
VDD5V	3.6	5.0	5.5	V
AVDD33	3.135	3.3	3.465	V
DVDD33	3.135	3.3	3.465	V
AVDD12	1.14	1.2	1.26	V
DVDD12	1.14	1.2	1.26	V
Input low voltage	GND	-	0.2 x DVDD33	V
Input high voltage	0.7 x DVDD33	-	DVDD33	V

Unless otherwise stated: ground = 0 V, temperature = +25 C.

8. Power Consumption and Converter Characteristics

8.1 Power by Mode

Mode	CVBS + audio	S-Video + audio	Test condition
Normal operation	0.4 W	0.5 W	Video/audio preview
Inactive	0.25 W	0.25 W	Video/audio off
Sleep	0.05 W	0.05 W	USB suspend

8.2 Typical Current by Module

Fixed output: 720 x 480 at 25 Hz, MJPEG. Typical values at ground = 0 V and +25 C.

Rail	CVBS + audio	S-Video + audio	Unit
DVDD12	27	27	mA
AVDD12	29	44	mA
AVDD33	14	27	mA
AVDD33_USB	36	36	mA
AVDD33_AU	11	11	mA
DVDD33	1	1	mA

8.3 Audio ADC

Parameter	Typical value	Unit
Resolution	24	bits
Channels	2	-
Dynamic range (SNR)	92	dB
AVDD33_AU supply current	11	mA

8.4 Video ADC

Parameter	Typical value	Unit
Resolution	10	bits
Channels	4	-
Dynamic range (SNR)	53	dB
AVDD33 supply current, normal mode	27	mA
AVDD12 supply current, normal mode	44	mA

9. Package Marking and Revision Record

The original document identifies a 7 mm x 7 mm LQFP-48 package. It depicts the standard top-view package outline but does not provide a full mechanical-dimension table in the extracted content.

Chip marking

The package marking contains the MS2106E product model, the manufacturer trademark, and a production-information code represented as XXXXXXXXXXXYY.

Revision history

Date	Version	Author	Description
January 2021	V1.0	MJ Du	Initial release

OpenXTCM5 schematic checklist

- Treat VDD5V as a switched, current-limited 5 V peripheral load, not a direct vehicle supply.
- Place USB ESD protection at the external/host boundary and route D+/D- as a 90-ohm differential pair.
- Keep the 12 MHz crystal, its loading network, and REXT close to the IC.
- Fit the external EEPROM footprint and bring GPIO2/3/4 out as the documented default configuration interface.
- In a video-only design, leave the audio path deliberately omitted and record the GPIO0 strap choice in the schematic.
- Bench-test USB enumeration, advertised UVC modes, actual latency, and temperature before committing to the MS2106E route.